

Day 2 : Mini project – Subset Selection Problem

1 .Problem statement :

- Find all the subsets from a set of numbers whose sum is zero.
- **Constraint:** Subset size must be 5
- Set={-12, -3, -6, 7, 2, -2, 6, 3, 9, -7, -5, -8, 1, 11, -9, -4}

Code :

```
from itertools import combinations

def find_zero_sum_subsets(input_set):
    subsets = list(combinations(input_set, 5))
    zero_sum_subsets = [subset for subset in subsets if sum(subset) == 0]
    return zero_sum_subsets

input_set = {-12, -3, -6, 7, 2, -2, 6, 3, 9, -7, -5, -8, 1, 11, -9, -4}
zero_sum_subsets = find_zero_sum_subsets(input_set)
for subset in zero_sum_subsets:
    print(subset)
```

Output :

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(1, 2, 3, 6, -12)
(1, 2, 3, -4, -2)
(1, 2, 6, -7, -2)
(1, 2, 6, -6, -3)
(1, 2, 6, -5, -4)
(1, 2, 7, -8, -2)
(1, 2, 7, -7, -3)
(1, 2, 7, -6, -4)
(1, 2, 9, -9, -3)
(1, 2, 9, -8, -4)
(1, 2, 9, -7, -5)
(1, 2, 11, -12, -2)
(1, 2, 11, -9, -5)
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(1, 2, 11, -8, -6)
(1, 3, 6, -8, -2)
(1, 3, 6, -7, -3)
(1, 3, 6, -6, -4)
(1, 3, 7, -9, -2)
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(1, 9, 11, -12, -9)
(1, 9, -5, -3, -2)
(1, 11, -7, -3, -2)
(1, 11, -6, -4, -2)
(1, 11, -5, -4, -3)
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(3, 9, -7, -3, -2)

(3, 9, -6, -4, -2)
(3, 9, -5, -4, -3)
(3, 11, -9, -3, -2)
(3, 11, -8, -4, -2)
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(7, 11, -8, -6, -4)
(7, 11, -7, -6, -5)
(9, 11, -12, -6, -2)
(9, 11, -12, -5, -3)
(9, 11, -9, -8, -3)
(9, 11, -9, -7, -4)
(9, 11, -9, -6, -5)
(9, 11, -8, -7, -5)

2.Problem statement :

- Find all the subsets from a set of numbers whose sum is zero.
- **Constraint:** Subset size must be 3 to 6 only
- Set={-12, -3, -6, 7, 2, -2, 6, 3, 9, -7, -5, -8, 1, 11, -9, -4}

Code :

```
from itertools import combinations

def find_zero_sum_subsets(input_set):
    zero_sum_subsets = []
    # Check combinations of size 3, 4, 5, and 6
    for subset_size in range(3, 7):
        subsets = combinations(input_set, subset_size)
        zero_sum_subsets.extend([subset for subset in subsets if sum(subset) == 0])
    return zero_sum_subsets

input_set = {-12, -3, -6, 7, 2, -2, 6, 3, 9, -7, -5, -8, 1, 11, -9, -4}
zero_sum_subsets = find_zero_sum_subsets(input_set)
for subset in zero_sum_subsets:
    print(subset)
```

Output :

(1, 2, -3)

(1, 3, -4)

(1, 6, -7)

(1, 7, -8)

(1, 11, -12)

(2, 3, -5)

(2, 6, -8)

(2, 7, -9)

(3, 6, -9)

(3, 9, -12)

(6, -4, -2)

(7, -5, -2)

(7, -4, -3)

(9, -7, -2)

(9, -6, -3)

(9, -5, -4)

(11, -9, -2)

(11, -8, -3)

(11, -7, -4)

(11, -6, -5)

(1, 2, 3, -6)

(1, 2, 6, -9)

(1, 2, 9, -12)

(1, 6, -5, -2)

(1, 6, -4, -3)

(1, 7, -6, -2)

(1, 7, -5, -3)

(1, 9, -8, -2)

(1, 9, -7, -3)

(1, 9, -6, -4)

(1, 11, -9, -3)

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(2, 3, 7, -12)
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(1, 2, 3, 6, -7, -5)
(1, 2, 3, 7, -9, -4)
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(1, 2, 3, 11, -12, -5)
(1, 2, 3, 11, -9, -8)
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(2, 6, 11, -9, -8, -2)
(2, 6, 11, -9, -7, -3)
(2, 6, 11, -9, -6, -4)
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